Duc Tam Nguyen

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EDUCATION

University of Illinois Chicago **Aug 2023 - May 2027**
B.S. in Computer Science, Minor in Mathematics *Chicago, IL*
- **GPA:** 3.6/4.0
- **Achievements:** Dean's List
- **Coursework:** Machine Learning, Applied Statistics, Computer Vision, Data Science, Big Data Mining

EXPERIENCES

Viettel AI (Viettel Group) **May 2025 - Aug 2025**
Computer Vision / AI Intern *Hanoi, Vietnam*
- Implemented and evaluated face recognition pipelines using ResNet and Vision Transformer (ViT) backbones in Python, benchmarking recognition accuracy and inference latency to assess model performance and deployment trade-offs.
- Analyzed NIST FRTE 1:N identification benchmarks across galleries of up to 1.6M identities, examining false positive and false negative identification rates (FPIR/FNIR) to evaluate accuracy and scalability.
- Synthesized experimental findings, technical benchmarks, and implementation constraints into comparative reports supporting internal computer vision research.

Viettel High Tech (Viettel Group) **May 2024 - Aug 2024**
Data Science Intern *Hanoi, Vietnam*
- Developed customer segmentation and propensity models using Python and SQL on multi-million-row telecom datasets, improving campaign targeting accuracy by 15%.
- Built and automated data cleaning, transformation, and feature engineering workflows using Python and SQL, reducing recurring reporting time by 30%.
- Conducted exploratory data analysis and data quality checks to identify customer behavior patterns, validate model inputs, and support data-driven marketing decisions.

PROJECTS

QuickScout Football Data Pipeline
- Built a quota-aware API-Football ELT pipeline, ingesting 20 player profiles and 20 season-stat records into PostgreSQL with pagination, bounded retries, deduplication, and idempotent upserts.
- Developed a 6-model dbt star schema with 76 automated CI data-quality tests; preserved 100% of 40 hosted season-stat rows across two idempotent builds with zero duplicate fact keys.
- Orchestrated a 6-task Airflow DAG for source validation, dbt transformations, and checksum verification; demonstrated retry recovery and consistent results across two local runs with zero duplicate records.

Repeat-Order Recommendation System
- Built an end-to-end repeat-order recommendation pipeline using Python, pandas, and scikit-learn, processing 1M+ retail transactions and generating 3.86M candidate recommendations.
- Engineered customer-product features and benchmarked heuristic, Logistic Regression, and XGBoost ranking models, achieving 0.378 NDCG@10 and 49.18% Recall@100 in candidate retrieval.
- Developed a FastAPI inference service with automated testing and performance benchmarking, achieving 12.53 ms p95 latency under 5 concurrent clients and 625 requests/sec throughput.

Text-Prompted Video Object Tracking
- Developed a text-prompted video object tracking pipeline integrating Grounding DINO for object detection and SAM2 for temporal mask propagation.
- Designed and evaluated baseline tracking and two re-grounding strategies across 19 video clips, identifying improvements in 9 clips with selective periodic re-grounding and analyzing temporal consistency trade-offs.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript
Data Engineering: PostgreSQL, dbt, Airflow, ETL/ELT, REST APIs, Data Modeling, Data Quality Testing
Cloud & DevOps: Google Cloud Run, Google Cloud Monitoring, Supabase, Vercel, Docker, Github Actions, CI/CD
Machine Learning & Analytics: scikit-learn, XGBoost, PyTorch, FastAPI, pandas

TEACHING EXPERIENCE

University of Illinois Chicago **Aug 2024 - Present**
Undergraduate Teaching Assistant, CS 151 & CS 341
- Supported 100+ students in discrete mathematics and programming language concepts through office hours, technical instruction, assignment grading, and collaboration with faculty.